Department of Electronic & Telecommunication Engineering
University of Moratuwa



EN4431 - Analog IC Design

Assignment 01
Voltage Controlled Oscillator (Ring vs. LC Tank) Analysis, Design, and Simulation

Nethmal W D I - 210421E

1 Theoretical Analysis

1.1 Ring VCO Analysis

Q1.1.A Barkhausen Criterion and Ring Oscillator Oscillation Condition

For a feedback circuit to sustain in oscillation at an angular frequency ω_0,

$$|A(j\omega_0)\beta(j\omega_0)| = 1 \tag{1}$$

$$\angle A(j\omega_0)\beta(j\omega_0) = 2\pi m, \qquad m \in Z \tag{2}$$

That is, the loop gain magnitude must be unity and the total loop phase shift must be an integer multiple of 360°. This is known as the Barkhausen Criterion. Consider a ring oscillator consists of N identical inverting delay stages connected in a loop, where each inverter contributes a phase inversion of 180° with a phase lag due to its finite delay ϕ_d, then the total loop phase shift is

$$\Phi_{loop} = N(180^\circ + \phi_d) \tag{3}$$

For sustained oscillation, this must satisfy

$$N(180^\circ + \phi_d) = 360^\circ m \tag{4}$$

Since each inverter contributes 180°, an odd number of stages is required so that the net inversion of the chain is effectively non inverted. The remaining 180° required for oscillation is then provided by the propagation delay of the stages at the oscillation frequency. Thus, for an odd-stage ring oscillator flushleft

For the gain condition. If each stage has small-signal magnitude response $|H_s(j\omega_0)|$, then the total loop-gain magnitude is

$$|H_{loop}(j\omega_0)| = |H_s(j\omega_0)|^N \tag{5}$$

At startup, oscillation requires flushleft

$$|H_s(j\omega_0)|^N \geq 1 \tag{6}$$

and at steady state, nonlinearities reduce the gain so that (6)

$$|H_s(j\omega_0)|^N = 1 \tag{7}$$

Therefore, an odd number of identical delay stages satisfies the Barkhausen criterion because the loop phase condition is met through inversion plus delay, and the gain condition is met when the product of the stage gains around the loop is unity at the oscillation frequency.

Q1.1.B Oscillation Frequency of a Ring VCO

A ring oscillator consists of N identical inverting stages, each having propagation delay t_d. When a transition occurs at the output of one stage, it must propagate through all N stages before returning to the same node. This requires a time of

$$Nt_d \tag{8}$$

This corresponds to only half of a complete oscillation cycle Therefore, the oscillation period is

$$T = 2Nt_d \tag{9}$$

$$f_{osc} = \frac{1}{T} = \frac{1}{2Nt_d} \tag{10}$$

When we consider the rise and fall propagation delays,

$$t_d = \frac{t_{PHL} + t_{PLH}}{2} \tag{11}$$

$$f_{osc} = \frac{1}{N(t_{PHL} + t_{PLH})} \tag{12}$$

1.2 Q1.1.C Relationship Between Control Voltage and Delay in a Current-Starved Inverter

The delay of one stage can be approximated as

$$t_d \approx \frac{Q}{I} = \frac{C_L \Delta V}{I_{ctrl}} \tag{13}$$

where:

- C_L is the effective node capacitance,
- ΔV is the voltage swing (here we use V_{DD}),
- I_{ctrl} is the current allowed by the current-starving transistor.

Hence,

$$t_d \approx \frac{C_L V_{DD}}{I_{ctrl}} \tag{14}$$

Considering that the starving transistor operates in saturation,

$$I_{ctrl} = \frac{1}{2}\mu C_{ox}\frac{W}{L}(V_{ctrl} - V_{th})^2 \tag{15}$$

let,

$$\beta = \mu C_{ox}\frac{W}{L} \tag{16}$$

$$t_d \approx \frac{2C_L V_{DD}}{\beta(V_{ctrl} - V_{th})^2} \tag{17}$$

Q1.1.D Laplace-Domain Transfer Function of the VCO

The output angular frequency of a VCO is proportional to the control voltage, where K_{VCO} is the VCO gain in units of rad/s/V.

$$\omega_{out}(t) = K_{VCO} V_{ctrl}(t) \tag{18}$$

$$\omega_{out}(t) = \frac{d\phi_{out}(t)}{dt} \tag{19}$$

$$\frac{d\phi_{out}(t)}{dt} = K_{VCO} V_{ctrl}(t) \tag{20}$$

$$\phi_{out}(t) = \int K_{VCO} V_{ctrl}(t)\, dt \tag{21}$$

$$\Phi_{out}(s) = \frac{K_{VCO}}{s} V_{ctrl}(s) \tag{22}$$

According to the Laplace-domain transfer function of the VCO, it act as an integrator in the phase domain,

$$H_{VCO}(s) = \frac{\Phi_{out}(s)}{V_{ctrl}(s)} = \frac{K_{VCO}}{s} \tag{23}$$

1.3 LC Tank VCO Analysis

Q1.2.A Resonance Frequency of an Ideal and Practical LC Tank

An ideal LC tank consists of an inductor L_1 and capacitor C_1 connected in parallel. Oscillation occurs at the resonance frequency where the inductive and capacitive reactances cancel each other,

$$X_L = X_C \tag{24}$$

$$\omega L_1 = \frac{1}{\omega C_1} \tag{25}$$

$$\omega_0 = \frac{1}{\sqrt{L_1 C_1}} \tag{26}$$

$$f_0 = \frac{1}{2\pi\sqrt{L_1 C_1}} \tag{27}$$

In practical the inductor exhibits a finite series resistance R_s. The realistic LC tank is therefore modeled as an inductor L_1 in series with R_s, connected in parallel with the capacitor C_1. The equivalent impedance is given by:

$$Z_{eq}(s) = \frac{R_s + sL_1}{1 + sC_1R_s + s^2L_1C_1} \tag{28}$$

$$|Z_{eq}(j\omega)|^2 = \frac{R_s^2 + L_1^2\omega^2}{(1 - L_1C_1\omega^2)^2 + R_s^2C_1^2\omega^2} \tag{29}$$

Unlike the ideal case, the impedance does not go to infinity due to the presence of R_s, and the circuit exhibits a finite Q.

For analysis we convert the series combination (L_1, R_s) into an equivalent parallel combination (L_P, R_P). Considering the impedances:

$$sL_1 + R_s = \frac{R_P \cdot sL_P}{R_P + sL_P} \tag{30}$$

Assuming steady-state operation,

$$L_P = L_1\left(1 + \frac{R_s^2}{L_1^2\omega^2}\right) \tag{31}$$

$$R_P = \frac{L_1^2\omega^2}{R_s} \tag{32}$$

Using the definition of quality factor $Q = \frac{\omega L_1}{R_s}$, we obtain:

$$R_P = Q^2R_s \tag{33}$$

Q1.2.B Negative Resistance concept

An ideal LC tank would oscillate indefinitely because energy continuously exchanges between the electric field of the capacitor and the magnetic field of the inductor without loss. But in a practical oscillator, parasitic resistance and the active devices also introduce loss. These losses can be represented by an equivalent parallel resistance R_p across the LC tank as we discussed earlier. Because of this resistance, the oscillation amplitude decays unless energy is continuously supplied. To sustain oscillation, an active circuit is used to provide negative resistance, which injects energy into the tank and cancels the energy dissipated in R_p.
For a standard cross-coupled NMOS pair, the small-signal equivalent negative resistance presented to the tank is approximately

$$R_{neg} \approx -\frac{2}{g_m} \tag{34}$$

equally, the negative conductance is

$$G_{neg} \approx -\frac{g_m}{2} \tag{35}$$

The tank loss conductance is

$$G_p = \frac{1}{R_p} \tag{36}$$

$$|G_{neg}| \geq G_p \tag{37}$$

Therefore,

$$\frac{g_m}{2} \geq \frac{1}{R_p} \tag{38}$$

$$g_m \geq \frac{2}{R_p} \tag{39}$$

The startup condition can be written as

$$g_m R_p \geq 2 \tag{40}$$

Hence, the cross-coupled NMOS pair provide transconductance so that its negative resistance cancels the tank loss resistance, to maintain a sustaining oscillation.

Q1.3 Comparison: Ring VCO vs LC VCO

Parameter	**Ring VCO**	**LC VCO**
Area	Small since no inductor	Large due to inductor
Phase Noise	High	Low
Tuning Range	Wide	Narrow
Power Consumption	Generally low	Generally high

Table 1: Comparison between Ring VCO and LC VCO

2 Design and Simulation

Justification for Choosing $N = 5$ Stages

The oscillation frequency of a ring oscillator is given by

$$f_{osc} = \frac{1}{2Nt_d} \tag{41}$$

For a target oscillation frequency of $100\,MHz$, the required delay per stage is

$$t_d = \frac{1}{2Nf_{osc}} \tag{42}$$

The choice of N directly impacts the required stage delay and overall oscillator performance. $N = 5$ is selected as an optimal trade-off for the following reasons.

- **Frequency Design Feasibility:** For $N = 5$ and $f_{osc} = 100\,MHz$, the required stage delay is approximately $1\,ns$, and it is practically achievable with current-starved CMOS inverters. In contrast, $N = 3$ would require a larger delay per stage, which may demand very low current and can make the circuit difficult to control and less stable.
- **Stability:** Increasing the number of stages improves the distribution of phase shift across the loop, making the oscillator less sensitive to variations in a single stage and improving robustness against process and supply variations.
- **Power and Area Trade-off:** While increasing N increases power consumption and area due to more stages switching, $N = 5$ provides a good balance. Using $N = 7$ or higher would unnecessarily increase power and area without significant performance improvement for this frequency range.

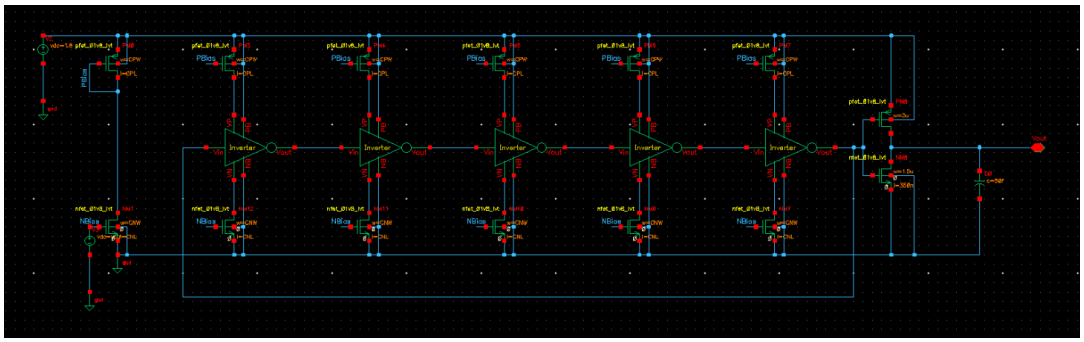


Figure 1: Initial Design

Transistor Sizes

Transistor	W	L
starved source NMOS	$750\,nm$	$350\,nm$
starved source PMOS	$1.5\,um$	$350\,nm$
INV NMOS	$3\,um$	$350\,nm$
INV NMOS	$6\,um$	$350\,nm$

Table 2: Transistor dimensions used in the VCO implementation

Q2.2.A

0.9V used as the nominal midpoint voltage and the 99.8M of mid frequency was achieved.

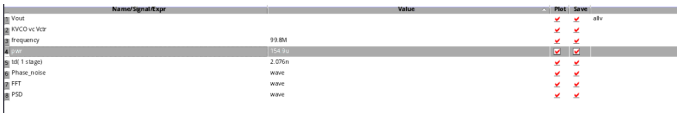


Figure 2: Testbench result

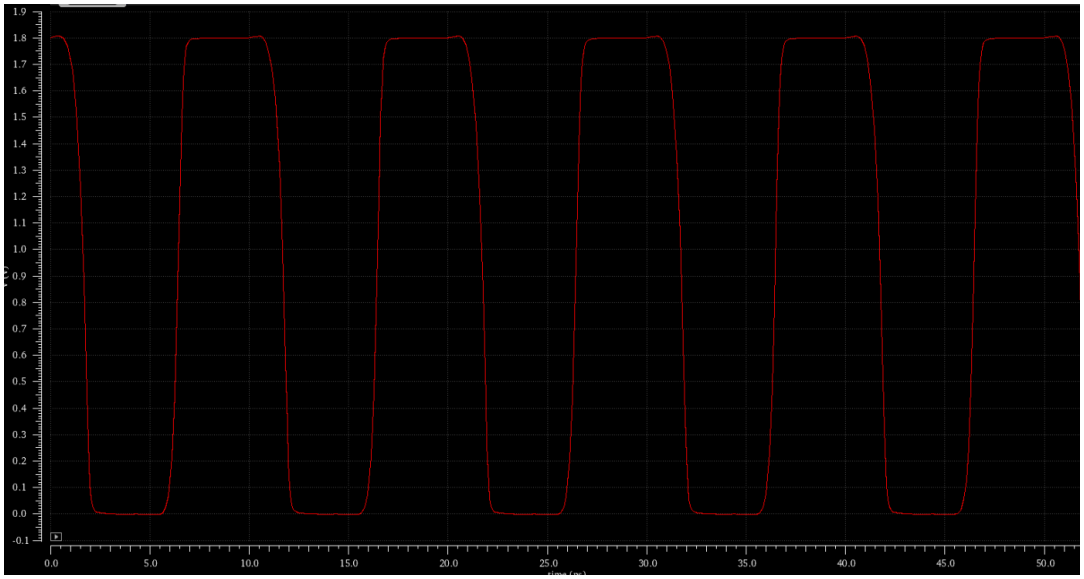


Figure 3: Transient output

Q2.2.B

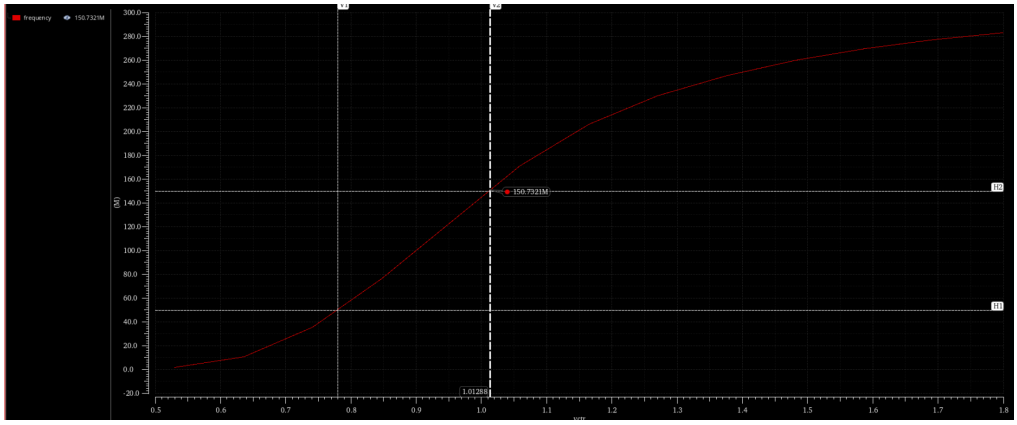


Figure 4: Fosc vs Vctr plot

Q2.2.C

Using the linear region of the simulated tuning curve, the VCO gain is calculated as the slope of the curve:

$$K_{VCO} = \frac{\Delta f_{osc}}{\Delta V_{ctrl}} = \frac{f_2 - f_1}{V_2 - V_1} \qquad (Hz/V) \tag{43}$$

$$K_{VCO,rad} = 2\pi K_{VCO,Hz} \tag{44}$$

$$K_{VCO} = \frac{150 - 50}{1.012 - 0.777} \qquad (Hz/V) \tag{45}$$

$$K_{VCO,rad} = 2\pi.425.532 = 2.6737 * 10^9 rad/s/V \tag{46}$$

Q2.2.D

The average power dissipation of the core oscillator at the 100MHz center frequency is around $154.9uW$.

Final Design

An improved resistor-controlled design was adopted form[1] to increase the linearity of the design, as shown below. This approach Here, the variation of VGS relative to VTH is small, resulting in a more linear change in current.

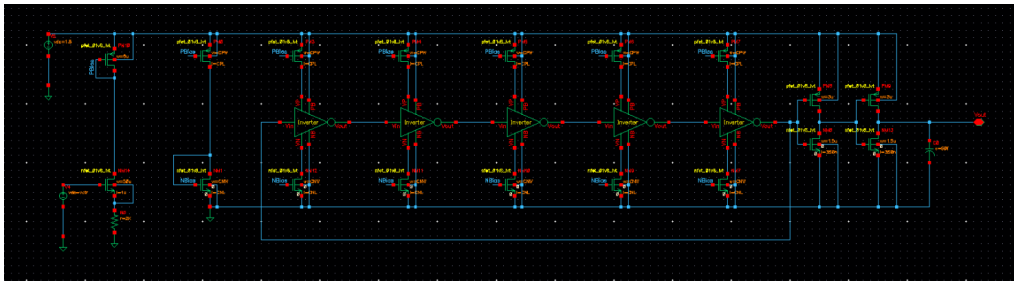


Figure 5: Final schematic

Q2.2.A

0.9V used as the nominal midpoint voltage and the 101.4M of mid frequency was achieved.

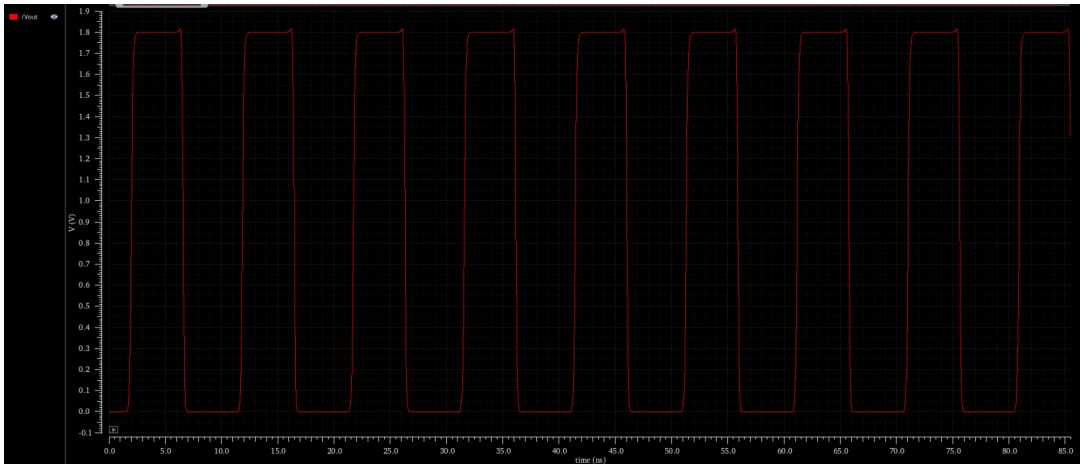


Figure 6: Transient output

Q2.2.B,C

$$K_{VCO} = \frac{168.9 - 36.9}{1.21 - 0.604} \qquad (Hz/V) \tag{47}$$

$$K_{VCO,rad} = 2\pi.217.822 = 1.369 * 10^9 rad/s/V \tag{48}$$

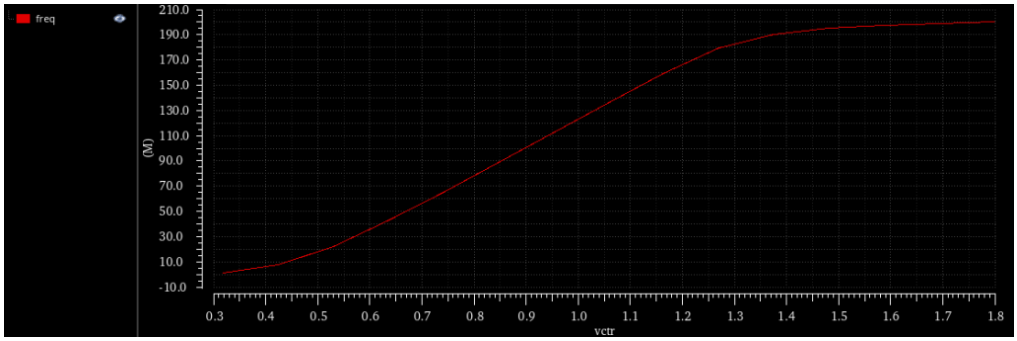


Figure 7: Fosc vs Vctr plot

Q2.2.D

The average power dissipation of the core oscillator at the 100MHz center frequency is around $360.1uW$.

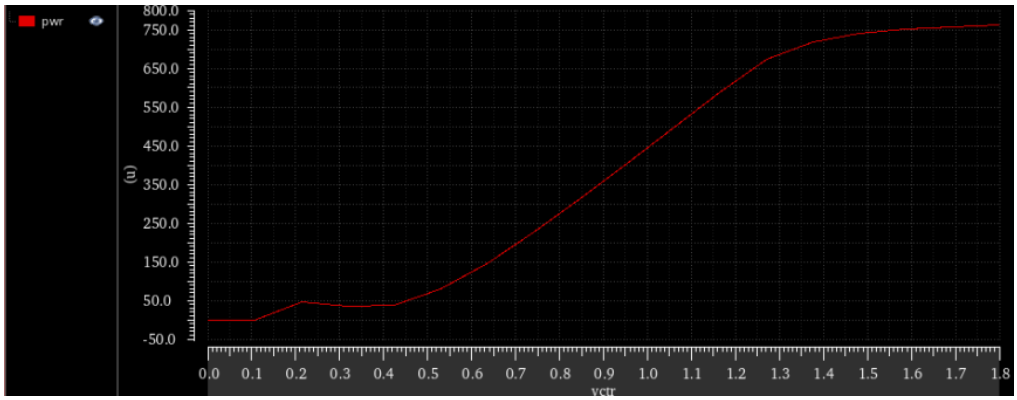


Figure 8: PWR vs Vctr

Q2.2.E Measured Performance Metrics of the VCO Design

- Supply voltage: $V_{DD} = 1.8\,V$
- Number of stages: $N = 5$
- Control-voltage range: 0 to $1.8\,V$
- Center frequency at $V_{ctrl} = 0.9\,V$: f_c= 101.4M
- Tuning range: 132
- VCO gain: 1369 Mrad/s/V
- Average power dissipation at center frequency: P_{avg} = 360.1uW

Q2.2.F Finalized Linear Transfer Function of the Simulated VCO

$$H_{VCO}(s) = \frac{\Phi_{out}(s)}{V_{ctrl}(s)} = \frac{K_{VCO}}{s} = \frac{1369Mrad/s/V}{s} \tag{49}$$

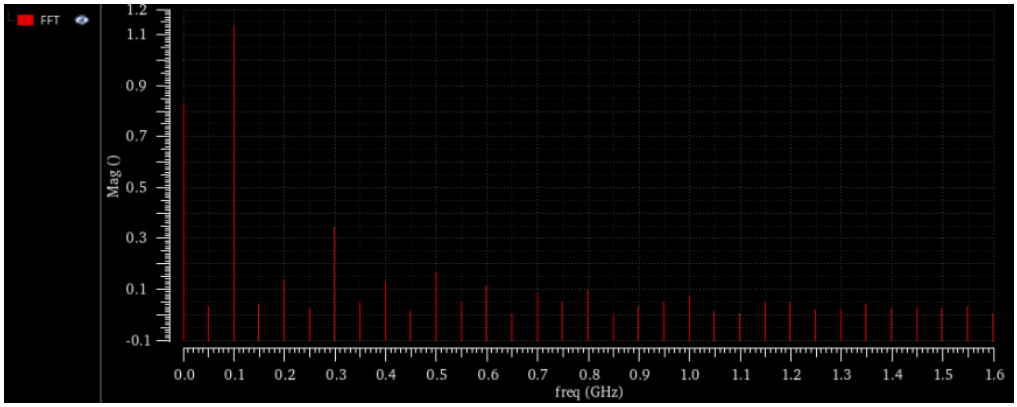


Figure 9: Frequency spectrum of the output

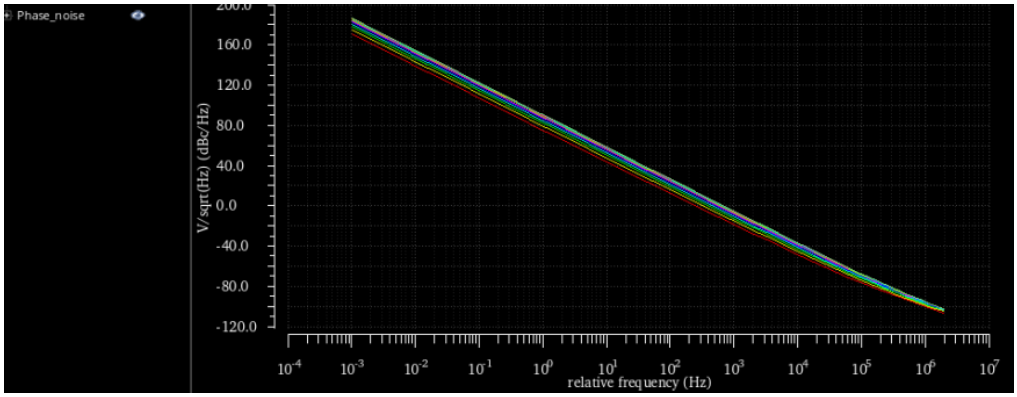


Figure 10: Phase noise

Improving the tuning range and stability of the VCO

1. Linearization using resistive biasing

A major limitation of current-starved VCOs is the nonlinearity between the control voltage V_{ctrl} and the oscillation frequency. This arises because the drain current of the MOS transistor depends nonlinearly on V_{ctrl},

$$I_D \propto (V_{ctrl} - V_{th})^2 \tag{50}$$

This leads to a nonlinear tuning curve. As discussed in reffered books's design approach, a linearization technique can be applied by introducing additional biasing elements to shape the control current.

By modifying the bias network, the current becomes approximately proportional to V_{ctrl},

$$I_D \propto V_{ctrl} \tag{51}$$

Since the oscillation frequency is proportional to the current:

$$f_{osc} \propto I_D \tag{52}$$

this results in a more linear relationship:

$$f_{osc} \propto V_{ctrl} \tag{53}$$

This improves:

- Linearity of the tuning curve
- Stability of K_{VCO}
- Predictability in PLL applications

2. Increasing effective output resistance of current source

This can be done by:

- Using longer channel lengths for the starving transistors
- By using cascode current sources

Increasing the output resistance reduces channel-length modulation, making the current source more ideal.

References

[1] R. Jacob Baker, CMOS: Circuit Design, Layout, and Simulation, 3rd ed., Wiley, 2010.

Appendix

```
// Generated for: spectre
// Generated on: Apr 12 22:09:05 2026
// Design library name: PLL_421E
// Design cell name: Linear_VCRO
// Design view name: schematic
simulator lang=spectre
global 0
parameters vctr=0.9 _gpar3=1 CNL=350n CNW=750n CPL=350n CPW=1.5u INL=350n \
    INW=1.15u IPL=350n IPW=2.3u nf=1 nfn=1
include "/home/aed/libs/Sky130_cadance_pdk/sky130_release_0.1.0/models/sky130.lib.spice"
include "/home/aed/libs/Sky130_cadance_pdk/sky130_release_0.1.0/models/corners/tt.spice"

// Library name: PLL_421E
// Cell name: inverter
// View name: schematic
subckt inverter NB PB VN VP Vin Vout
    PM0 (Vout Vin VP PB) pfet_01v8_lvt w=(IPW/nf) l=IPL as=145.75f \
        ad=145.75f ps=1.63u pd=1.63u m=(1)*(nf)
    NM0 (Vout Vin VN NB) nfet_01v8_lvt w=(INW/nfn) l=INL as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(nfn)
ends inverter
// End of subcircuit definition.

// Library name: PLL_421E
// Cell name: Linear_VCRO
// View name: schematic
NM14 (PBias net048 net030 net030) nfet_01v8_lvt w=(50u) l=1u as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
NM13 (Vout net018 0 0) nfet_01v8_lvt w=(1.5u) l=350n as=111.3f ad=111.3f \
        ps=1.37u pd=1.37u m=(1)*(1)
NM12 (net14 NBias 0 0) nfet_01v8_lvt w=(CNW/_gpar3) l=CNL as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(_gpar3)
NM8 (net018 net015 0 0) nfet_01v8_lvt w=(1.5u) l=350n as=111.3f ad=111.3f \
        ps=1.37u pd=1.37u m=(1)*(1)
NM7 (net021 NBias 0 0) nfet_01v8_lvt w=(CNW/_gpar3) l=CNL as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(_gpar3)
NM11 (net13 NBias 0 0) nfet_01v8_lvt w=(CNW/_gpar3) l=CNL as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(_gpar3)
NM10 (net027 NBias 0 0) nfet_01v8_lvt w=(CNW/_gpar3) l=CNL as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(_gpar3)
NM1 (NBias NBias 0 0) nfet_01v8_lvt w=(CNW/_gpar3) l=CNL as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(_gpar3)
NM9 (net024 NBias 0 0) nfet_01v8_lvt w=(CNW/_gpar3) l=CNL as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(_gpar3)
I10 (0 net3 net021 net020 net022 net015) inverter
I9 (0 net3 net024 net025 net023 net022) inverter
I8 (0 net3 net027 net026 net028 net023) inverter
I6 (0 net3 net14 net16 net015 net029) inverter
I7 (0 net3 net13 net15 net029 net028) inverter
PM10 (PBias PBias net3 net3) pfet_01v8_lvt w=(6u) l=350n as=145.75f \
        ad=145.75f ps=1.63u pd=1.63u m=(1)*(1)
```

```
PM9 (Vout net018 net3 net3) pfet_01v8_lvt w=(3u) l=350n as=145.75f \
        ad=145.75f ps=1.63u pd=1.63u m=(1)*(1)
PM8 (net018 net015 net3 net3) pfet_01v8_lvt w=(3u) l=350n as=145.75f \
        ad=145.75f ps=1.63u pd=1.63u m=(1)*(1)
PM7 (net020 PBias net3 net3) pfet_01v8_lvt w=(550n) l=CPL as=145.75f \
        ad=145.75f ps=1.63u pd=1.63u m=(1)*(1)
PM6 (net025 PBias net3 net3) pfet_01v8_lvt w=(550n) l=CPL as=145.75f \
        ad=145.75f ps=1.63u pd=1.63u m=(1)*(1)
PM5 (net026 PBias net3 net3) pfet_01v8_lvt w=(550n) l=CPL as=145.75f \
        ad=145.75f ps=1.63u pd=1.63u m=(1)*(1)
PM3 (net16 PBias net3 net3) pfet_01v8_lvt w=(550n) l=CPL as=145.75f \
        ad=145.75f ps=1.63u pd=1.63u m=(1)*(1)
PM0 (NBias PBias net3 net3) pfet_01v8_lvt w=(550n) l=CPL as=145.75f \
        ad=145.75f ps=1.63u pd=1.63u m=(1)*(1)
PM4 (net15 PBias net3 net3) pfet_01v8_lvt w=(550n) l=CPL as=145.75f \
        ad=145.75f ps=1.63u pd=1.63u m=(1)*(1)
R0 (net030 0) resistor r=2K
V5 (net3 0) vsource dc=1.8 type=dc
V4 (net048 0) vsource dc=vctr type=dc
C0 (Vout 0) capacitor c=50f
ic net015=0
simulatorOptions options reltol=1e-3 vabstol=1e-6 iabstol=1e-12 temp=27 \
    tnom=27 scalem=1.0 scale=1.0 gmin=1e-12 rforce=1 maxnotes=5 maxwarns=5 \
    digits=5 cols=80 pivrel=1e-3 sensfile="../psf/sens.output" \
    checklimitdest=psf
tran tran stop=1u errpreset=moderate write="spectre.ic" \
    writefinal="spectre.fc" annotate=status maxiters=5
finalTimeOP info what=oppoint where=rawfile
pss ( Vout 0 ) pss fund=101M harms=30 errpreset=conservative
+    tstab=15n annotate=status
pnoise ( Vout 0 ) pnoise sweeptype=relative relharmnum=1
+       start=1m stop=2M pnoisemethod=fullspectrum noisetype=timeaverage
+       noiseout=[am pm usb lsb] annotate=status
modelParameter info what=models where=rawfile
element info what=inst where=rawfile
outputParameter info what=output where=rawfile
designParamVals info what=parameters where=rawfile
primitives info what=primitives where=rawfile
subckts info what=subckts where=rawfile
save Vout
saveOptions options save=all pwr=all
```